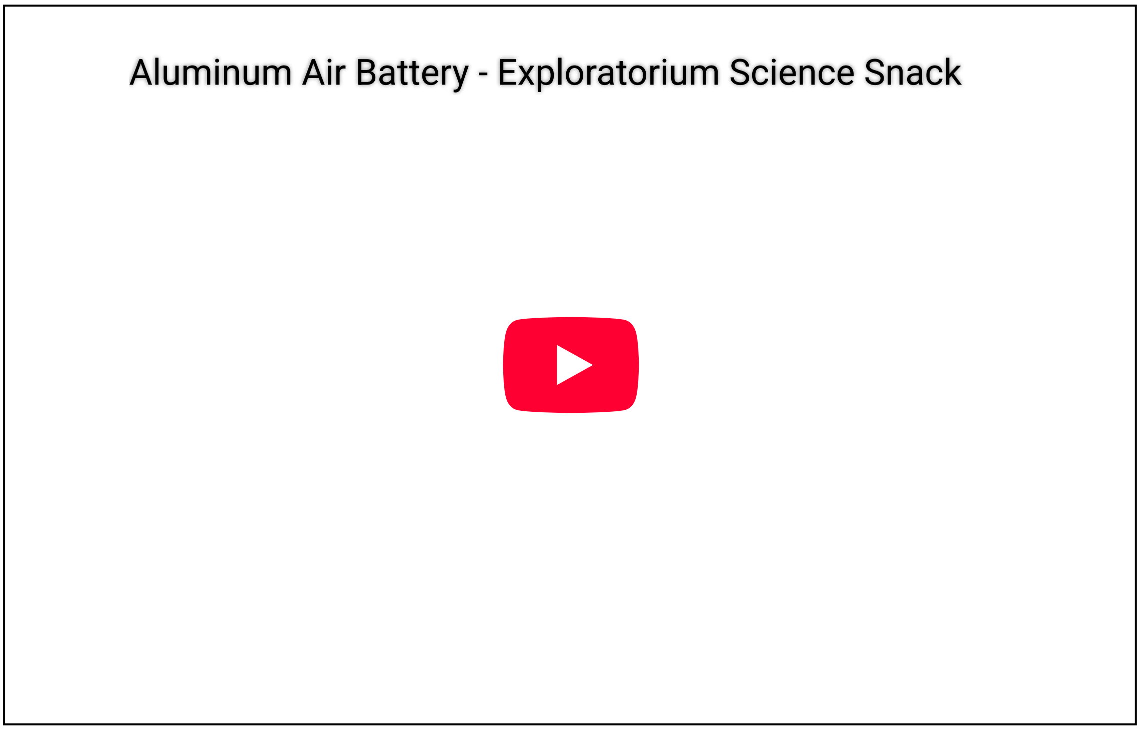
Voltaic cells and rechargeable batteries

Learning outcomes

By the end of this section you should be able to:

- (R3.2.6) Explain the direction of electron flow from anode to cathode in the external circuit, and ion movement across the salt bridge.
- (R3.2.7) Deduce the reactions of the charging process from given electrode reactions for discharge, and vice versa.

Batteries are, in essence, very simple devices. Look at the simple aluminium/air battery made of activated charcoal, aluminium foil and a salt solution in **Figure 1**.



Video 1. An aluminium/air battery powering up a lamp.

Eventually batteries ‘die’ and need to be replaced – what do you think is happening in terms of the chemical reaction inside?

Creativity, activity, service

Strand: Service

Learning outcomes:

- Demonstrate engagement with issues of global significance
- Recognise and consider the ethics of choices and actions

The use of electric cars is likely to increase over the next decade as governments aim to phase out the production (https://www.statista.com/chart/29359/official-targets-phase-out-sale-gasoline-cars/) of automobiles that run on petrol (gasoline) or diesel. At first, this might seem to be an environmental benefit due to the reduction in carbon dioxide emissions. However, an increase in the production of the batteries required for electric cars means that not all countries will feel this benefit. The materials required for the production of these batteries are often found in developing countries and according to the UNCTAD, it is these countries that may end up paying the greatest human and environmental cost (https://unctad.org/news/developing-countries-pay-environmental-cost-electric-car-batteries) of switching over to electric cars.

As an IB student, what can you do to raise awareness of such issues? Should we consider all aspects of our behaviour as consumers even if the repercussions do not directly affect us?

Voltaic cells (primary batteries)

In a voltaic cell, electricity is produced as a redox reaction proceeds. This means at the anode, electrons will be lost, and they will be gained at the cathode. If the transfer of electrons is done through a wire connecting both electrodes, the electricity created can be harnessed. To allow electrons to move in a wire, the two half-reactions need to be separated into two half-cells.

In metal/metal-ion electrode batteries, each half-cell consists of a bar of metal dipped into a solution containing cations of the same metal, like the one in **Figure 1**.

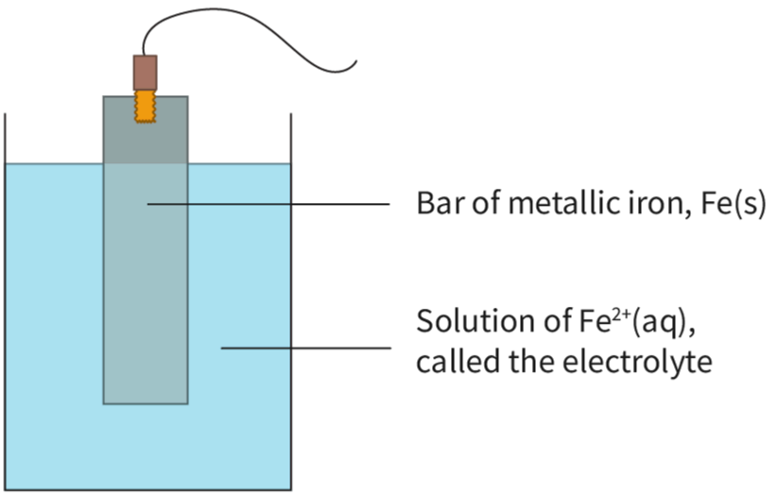


Figure 1. An example of a half-cell of Fe(s) with its metal ion in solution.

More information for figure 1

The image is a diagram depicting a half-cell used in metal/metal-ion electrode batteries. In the diagram, there is a beaker filled with a light blue liquid, labeled as a 'Solution of Fe^{2+}(aq), called the electrolyte.' Inside the beaker, a vertical bar labeled 'Bar of metallic iron, Fe(s)' is partially submerged in the solution. The bar is connected to a wire at the top, representing an electrode system. The diagram illustrates the setup of the iron in the electrolyte solution, which is typical for demonstrating electrochemical reactions in such half-cells.

[Generated by AI]

You see an electrode of iron, Fe(s), in the electrolyte, a solution of iron ions, Fe^{2+}(aq). The half-cell can be represented as:

$$Fe(s) \mid Fe^{2+}(aq)$$

The straight line between the metal and metal ion represents the phase boundary or junction, the surface of the metal in contact with the solution. In the half-cell, the atoms of the metal will form ions by releasing electrons. So, the metal surface is negative compared to the solution. This charge separation is called the electrode potential. The electrode potential denotes an equilibrium between the metal and its ions in the electrolyte.

You can have an electrode of any metal and its metal ion. Practice with these two examples that represent half-cells in voltaic cells.

In each voltaic cell there are two half-cells connected through the wire attached to the metal bars. In that wire, you can find a lamp or a voltmeter represented but they are not required for the running of the battery. A lamp, or any other device, can be placed on the wire to harness the electricity being produced by the electron flow, while a voltmeter is a device that simply reads the electrical potential created by the movement of electrons from the zinc to the copper half-cells. Voltmeters are used to indicate if the battery is working and how much energy is being produced. They should give a positive value as energy is being produced. If the value is negative it means the wires were attached to the incorrect poles. To fix this, you just need to reverse the cables attached to the anode and cathode.

Once the half-cells are connected by the external wire, the reaction can proceed, but will immediately run into a problem. As electrons flow from the anode to the cathode, there will be a build-up of negative charge at the cathode and positive charge at the anode, which makes the cell stop. In order to prevent this from happening, another component is required; the salt bridge. The salt bridge will allow the continuous flow of electrons on the external wire because it provides a liquid junction between the two half-cells with free to move ions. A salt bridge is essential for the functioning of a battery because it:

- physically separates the cathode and anode as there is no mixing of the two solutions.
- provides an electrical path for the movement of the cations and anions in the cell.
- reduces the liquid-junction potential (voltage generated when two different solutions come into contact with each other, which occurs due to unequal cation and anion movement across the junction).

The cell in **Interactive 1** needs your help to function. Despite the fact that the electrodes and wire are in place, the electrons are not flowing in the wire. You need to balance the charges accumulating at each half-cell with the ions from the salt bridge. Drag the correct boxes to the spaces available on the voltaic cell to restore the production of electricity in the battery.

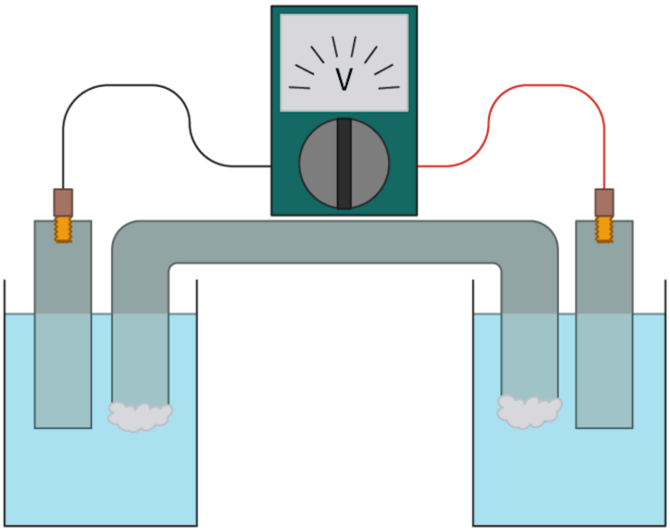
Interactive 1. A Voltaic Cell and the Movement of Ions on the Salt Bridge.

Interactive 1. A voltaic cell and the movement of ions on the salt bridge.

As you can see, the salt bridge contains a concentrated solution of a strong electrolyte which allows ions to diffuse out of it. The ions should be inert (not react with the other ions in the solution). Two common salt solutions in salt bridges are sodium sulfate (Na_2SO_4) and potassium chloride (KCl). At the openings of a salt bridge, there is generally cotton wool that prevents the solution from leaving the tube but still allows the movements of ions in and out.

In a salt bridge, the ions will move to the half-cell that is accumulating an opposite charge. In the example in **Figure 2** the positive ion in the salt bridge (K^+) is moving to the cathode as there has been an accumulation of negative charge (electrons arriving from the anode). The anion, Cl^-, moves to the anode as there has been an accumulation of the iron ions, Fe^{2+}.

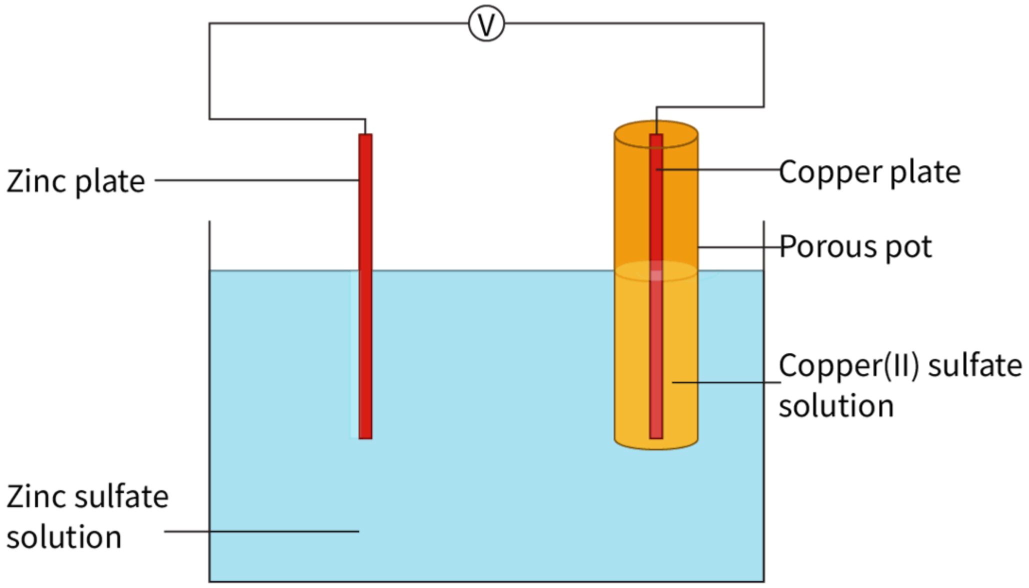
In a school setting, sometimes a porous cup or paper soaked in a salt solution is used as a simplified version of a salt bridge as in **Figure 2**.



More information

The image shows a diagram of an electrochemical cell setup. It includes two containers filled with a solution connected by a salt bridge. Each container has an electrode submerged in the solution. The electrodes are connected by wires to a voltmeter placed on top of the setup. The two half-cells are connected by a salt bridge.

[Generated by AI]



More information

The image is a diagram of a galvanic cell, depicting a simple electrochemical setup. On the left side, there is a zinc plate labeled "Zinc plate," immersed in a "Zinc sulfate solution." On the right side, there is a "Copper plate" placed inside a "Porous pot," which is immersed in a "Copper(II) sulfate solution." The zinc and copper plates are connected by a wire with a voltmeter symbolized by a circle marked with a "V." The wire indicates a flow of electrons from the zinc electrode to the copper electrode. The diagram illustrates the different components of the galvanic cell and their interactions, showing how chemical energy is converted into electrical energy.

[Generated by AI]

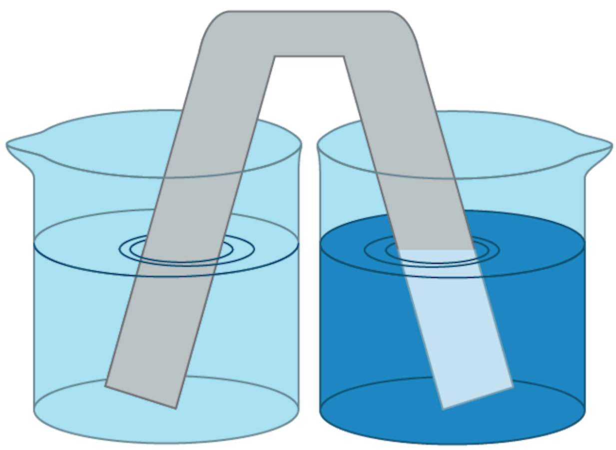


Figure 2. Examples of salt bridges used in a school laboratory setting.

More information for figure 2

The image shows an illustration of a laboratory setup involving two beakers, each partially filled with liquid. The beaker on the left contains a light blue liquid, while the beaker on the right holds a darker blue liquid. A gray, inverted U-shaped tube, representing a salt bridge, connects the two beakers. This setup is commonly used to demonstrate electrochemical cells where ions can pass through the salt bridge to complete the electrical circuit.

[Generated by AI]

In 1836, a British chemist and meteorologist called John Frederic Daniell invented a type of electrochemical cell that became known as the Daniell cell. In it, he improved the voltaic pile created by Alessandro Volta by separating the half-cells of copper and zinc by a porous ceramic cup, known as a 'semi-permeable diaphragm.' The electrodes used in the Daniell call can be represented as:

$Cu(s) \mid Cu^{2+}(aq)$ and $Zn(s) \mid Zn^{2+}(aq)$

Figure 3. A Daniell cell recreated in a school laboratory.

Source: "Daniell cell demonstration (https://commons.wikimedia.org/wiki/File:Daniell_cell_demonstration.jpg)" by Taweetham is licensed under CC BY-SA 4.0 (https://creativecommons.org/licenses/by-sa/4.0/deed.en)

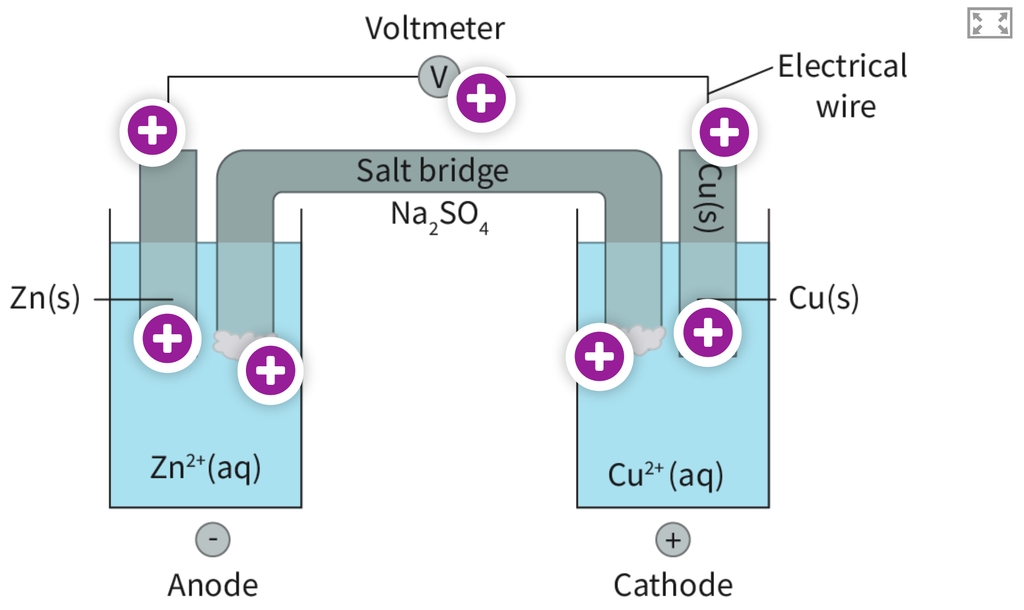
More information for figure 3

The image shows a recreated Daniell cell in a laboratory setting. There are two clear beakers. The left beaker is labeled "ZnSO4" and contains a clear liquid with a metallic strip connected to a wire. The right beaker is labeled "CuSO4" and is filled with a blue liquid, also with a metallic strip connected to a wire. These wires lead to a digital multimeter on the left, displaying a reading of 1.07 volts. The setup demonstrates the basic principles of electrochemistry with zinc and copper electrodes in their respective sulfate solutions, connected to measure voltage.

[Generated by AI]

In the Daniell cell in **Figure 3**, zinc is more easily oxidised than copper and undergoes oxidation. The equilibrium between metal and metal ions on the copper electrode is more towards the metal (less tendency to lose electrons) so copper has a higher electrode potential than zinc. This means that the zinc atoms are more likely to release electrons; the zinc half-cell is more negative. The copper ions are more likely to accept electrons; the copper half-cell is more positive. This results in the flow of electrons from the zinc to the copper half-cell.

Use the hotspots in **Interactive 2** to learn more about what is happening at each electrode, the electrical wire, and the salt bridge.



Interactive 2. What Happens Inside a Daniell Cell?

More information for interactive 2

An interactive displays a diagram of a galvanic cell with two half-cells connected by a salt bridge containing Na_2SO_4. The left half-cell contains a zinc electrode labeled Zn solid immersed in an aqueous solution of Zn^{2+} . The right half-cell contains a copper electrode labeled Cu solid immersed in an aqueous solution of Cu^{2+} . The zinc electrode is labeled anode with a negative sign, and the copper electrode is labeled cathode with a positive sign. A voltmeter is connected between the two electrodes through an electrical wire. White clouds are near the salt bridge openings in both half-cells.

There are 7 Hotspots represented by plus signs at various places. Hotspot 1 above the zinc electrode. Hotspot 2 near the voltmeter. Hotspot 3 above the copper electrode. Hotspot 4 on the zinc electrode. Hotspot 5 on the aqueous solution of zinc. Hotspot 5 on the aqueous solution of copper. Hotspot 7 on the copper electrode. Clicking on the hotspots reveals what is happening in that place.

The following items are revealed at respective hotspots:

Hotspot 1: The electrons lost by zinc are carried by the electrical wire in the direction of the cathode.
Hotspot 2: The voltmeter is the device that reads the electrical potential created by the movement of electrons from the zinc to the copper half-cells.
Hotspot 3: The electrons lost by zinc arrive at the cathode and will be received by the copper ions in the solution.
Hotspot 4: In the anode, you will see the zinc metal breaking apart as the zinc atoms will be losing electrons to the external wire. The zinc ions are added to the electrolyte in the zinc half-cell. The zinc bar gets thinner.
Hotspot 5: Due to the increase of positive charges in the zinc half-cell by the addition of zinc ions, the negative ions in the salt bridge (SO_4^{2-}) move to the solution to balance out the charges and allow the reaction to continue.
Hotspot 6: Due to the decrease of positive charges in the copper half-cell by the removal of copper ions, the positive ions in the salt bridge (Na^+) move to the solution to balance out the charges and allow the reaction to continue.
Hotspot 7: The copper ions approach the metal and receive the electrons that arrived on the electrical wire. Copper metal is created, and the copper bar increases. The blue colour of the copper ion solution fades due to the removal of the copper ions.

Nature of Science

Aspect: Experiments

Kog

Alessandro Volta, an Italian scientist, is credited with the invention of the first true voltaic cell, known as the 'voltaic pile,' in 1800. Volta conducted a series of experiments to investigate the effects of different metals and electrolytes on the generation of electricity. He stacked alternating layers of zinc and copper discs separated by pieces of cardboard soaked in salt water, which served as the electrolyte. However, the voltaic pile had limitations such as low voltage output, short lifespan, and the production of corrosive chemicals.

John Daniell, a contemporary of Volta, sought to improve upon these limitations. The Daniell cell used a porous ceramic cup, known as a 'semipermeable diaphragm,' to separate the copper sulfate solution from the zinc sulfate solution. The Daniell cell had several advantages: it had a more stable and longer-lasting output of electrical energy, a higher voltage output and produced fewer corrosive chemicals.

Over the years, extensive experimentation has been conducted in an attempt to develop more modern and efficient voltaic cells. For example, researchers have experimented with different electrode materials, electrolyte compositions, and cell configurations to optimise cell performance, durability, and safety. These experiments have led to the development of various types of voltaic cells, which are used in everyday devices, such as batteries, flashlights, and calculators and also more modern designs like fuel cells (see section R1.3.5 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/fuel-cells-id-44783/)) and solar cells that are used in renewable energy technologies today. Despite the benefits of producing cleaner renewable energy and increased efficiency, both fuel and solar cells are not only expensive solutions but have other drawbacks.

All these experiments show the tentative and dynamic nature of scientific knowledge, as new discoveries and technologies can challenge or modify existing theories and models and help solve new and old problems. They also reflect the influence of financial, social, and cultural factors on scientific practice and knowledge production.

A voltaic cell can be represented using a cell diagram as in **Figure 4**. The convention is to draw the anode on the left and the cathode on the right. Note that the phase boundary between a solid and an aqueous solution is represented by a single vertical line and the salt bridge is represented by the two parallel vertical lines between the anode and the cathode.

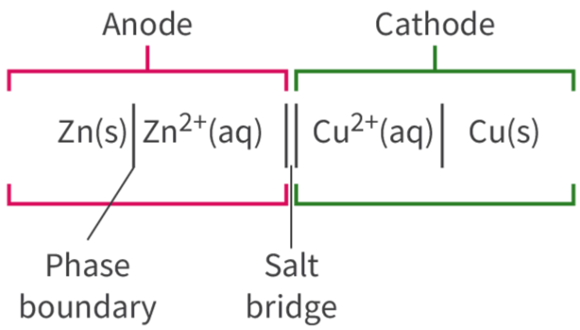


Figure 4. The cell diagram convention for a voltaic cell.

More information for figure 4

This diagram represents a voltaic cell with an anode on the left and a cathode on the right. The anode side features zinc in solid form (Zn(s)) transitioning to zinc ions in aqueous solution (Zn2+(aq)), indicated by a single vertical line representing the phase boundary. The cathode portion shows copper ions in aqueous solution (Cu2+(aq)) transitioning to copper in solid form (Cu(s)), separated by a phase boundary. Between these two sections is a salt bridge, symbolized by two parallel vertical lines. Labels indicate 'Anode' above the left section and 'Cathode' above the right section, with 'Phase boundary' and 'Salt bridge' labeled under their respective lines.

[Generated by AI]

In **Interactive 1**, write the cell diagram for a voltaic cell with electrodes of nickel and lead by dragging the boxes to the correct locations below.

$$Ni(s) + Pb^{2+}(aq) \rightarrow Ni^{2+}(aq) + Pb(s)$$

Pb(s)	Ni(s)	$Pb^{2+}(aq)$	$Ni^{2+}(aq)$	|	| |

Check

Interactive 3. A voltaic cell with electrodes of nickel and lead.

Study skills

Construction of primary cells should include: half-cells containing metal/metal ion, anode, cathode, electric circuit, and salt bridge, as in **Figure 5.**

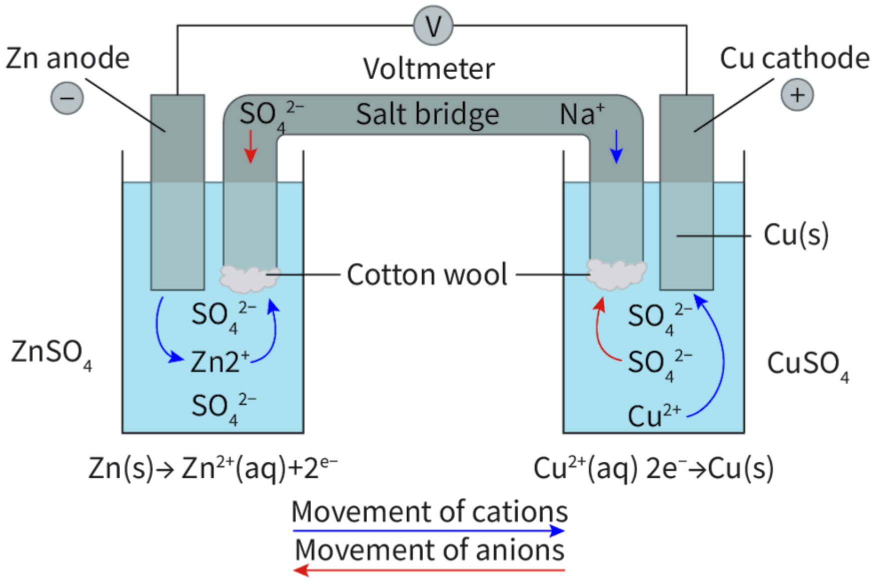


Figure 5. A representation of a voltaic cell.

More information for figure 5

This diagram illustrates a voltaic cell consisting of two half-cells connected by a salt bridge. On the left, there is a zinc (Zn) anode submerged in a $ZnSO_4$ solution. The zinc undergoes oxidation, releasing electrons and forming Zn^{2+} ions. The movement of cations is indicated by a blue arrow heading towards the salt bridge. Below the anode, the reaction is represented as $Zn(s) \rightarrow Zn^{2+}(aq) + 2e^{-}$.

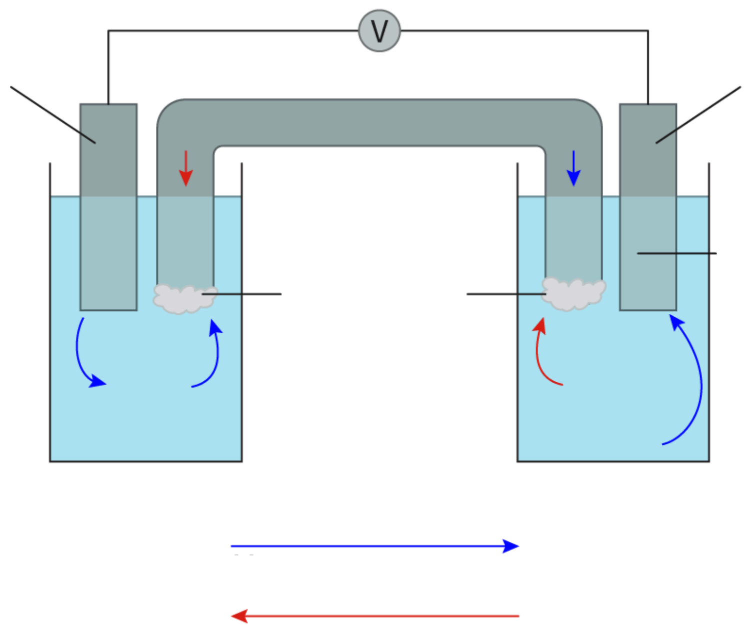
On the right, a copper (Cu) cathode is submerged in a $CuSO_4$ solution. Here, Cu^{2+} ions gain electrons to form copper metal. The movement of anions, indicated by red arrows, also heads towards the salt bridge. The cathodic reaction is depicted as $Cu^{2+}(aq) + 2e^{-} \rightarrow Cu(s)$.

A salt bridge, filled with Na^{+} and SO_4^{2-} ions, allows for the movement of ions between the two half-cells, maintaining electrical neutrality. Cotton wool appears to be used as a medium within the salt bridge. A voltmeter is connected between the anode and cathode to measure the potential difference.

Overall, the diagram depicts the flow of electrons from the anode to the cathode through an external circuit, while ions travel through the solutions and salt bridge to balance the charge.

[Generated by AI]

Use **Interactive 4** to practice labelling a voltaic cell.



Movement of cations	Ni anode	Cotton wool	Na^+	Ni^{2+}	+
Movement of anions	$Pb^{2+}(aq) + 2e^- \rightarrow Pb(s)$	Pb cathode	Pb^{2+}	$PbSO_4$	−
Salt bridge	$Ni(s) \rightarrow Ni^{2+}(aq) + 2e^-$	Voltmeter	SO_4^{2-}	$NiSO_4$	Pb(s)

Check

Interactive 4. Label the voltaic cell.

Rechargeable batteries (secondary batteries)

Unlike the single use batteries studied in the previous section, cell phones, computers and cars use rechargeable batteries. Many electric or electronic devices can be powered by rechargeable batteries.

Video 2. Many Devices Use Rechargeable Batteries.

More information for video 2

A video shows a close-up of a battery charger holding four cylindrical rechargeable batteries. The screen on the charger is illuminated with a bright blue backlight, displaying four battery icons. Each icon represents a battery's charging status, with three of them showing low charge levels and one showing a slightly higher charge, indicated by partially filled bars. The screen also displays the text CHG at the top left, signifying that the batteries are currently in the process of charging. The background is slightly blurred, focusing attention on the charging display.

How do these batteries compare to the ones you already studied?

How do rechargeable batteries work? Can you guess how redox reactions might occur in rechargeable batteries?

Rechargeable batteries use chemical reactions to provide energy to power devices, just like primary cells, but can be used more than once, if they are recharged. To recharge, the chemical reaction is reversed by the use of an electrical current to drive the reaction back to its original state. The current used needs to be carefully controlled to prevent damage to the battery. That is why generally there are specific chargers for specific batteries, as they monitor the current and temperature of the battery to extend battery life. Battery life is a measure of the time that the battery will continue to provide energy. This means that even with proper care and maintenance, rechargeable batteries will eventually need to be replaced.

The most common types of rechargeable batteries for daily use include nickel-metal hydride (NiMH), typical batteries like AA or AAA, and lithium-ion (Li-ion) batteries like the ones used in cell phones and laptops.

Let's look at what happens inside a lithium-ion battery made up of lithium cobalt oxide ($LiCoO_2$). In this type of cell, the cathode is made of a lithium metal oxide such as lithium cobalt oxide ($LiCoO_2$) and the anode is composed of lithium atoms absorbed onto a lattice of graphite. The electrolyte is a non-aqueous solution containing lithium ions, which flow between the cathode and anode during charging and discharging.

When being discharged, the following half equations take place. At the anode, the lithium atoms undergo oxidation to form lithium ions.

$$\text{Li(graphite)} \rightarrow \text{Li}^+\text{(electrolyte)} + e^-$$

The electrons flow in the external circuit from anode to cathode. The lithium ions flow through the electrolyte to the cathode where they undergo reduction.

$$\text{Li}^+\text{(electrolyte)} + e^- + \text{CoO}_2\text{(s)} \rightarrow \text{LiCoO}_2\text{(s)}$$

Note that when being recharged, the two half equations above are reversed. During charging, lithium ions flow from the cathode to the anode through the electrolyte. This process stores energy in the battery. During discharging, the lithium ions flow back from the anode to the cathode through the electrolyte, generating an electric current that can be used to power devices.

Note that these reactions are simplified representations of the actual reactions that occur in rechargeable batteries. The actual reactions involve multiple steps and intermediate compounds.

Learn more about other types of rechargeable batteries and predict their discharge or charge reactions with the examples below.

Worked example 1

A nickel-metal hydride (NiMH) battery is a very common AA or AAA battery. This is the reaction that happens during discharge:

$$MH + NiO(OH) + H_2O \rightarrow MH(OH) + Ni(OH)_2$$

State the charging reaction.

$$MH(OH) + Ni(OH)_2 \rightarrow MH + NiO(OH) + H_2O$$

To write the reverse equation, simply swap the reactants and products.

The reaction that occurs during discharge is the oxidation of the metal hydride and the reduction of nickel oxide hydroxide.

During charging, the reaction is reversed, and nickel hydroxide is oxidised while the metal hydride is reduced.

Worked example 2

Nickel-cadmium (NiCd) batteries were commonly used in cordless power tools such as drills, saws, and screwdrivers, but have mostly now been replaced.

The reaction that happens during discharge is:

$$Cd + NiO_2 + 2H_2O \rightarrow Cd(OH)_2 + Ni(OH)_2$$

State the charging reaction.

$$Cd(OH)_2 + Ni(OH)_2 \rightarrow Cd + NiO_2 + 2H_2O$$

To write the reverse equation, simply swap the reactants and products.

The reaction that occurs during discharge is the oxidation of the cadmium and the reduction of nickel oxide.

During charging, the reaction is reversed, and nickel hydroxide is oxidised while the cadmium hydroxide is reduced.

Making connections

Secondary cells rely on electrode reactions that are reversible. What are the common features of these reactions? (see subtopic R2.3 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43482/)).

Reversible reactions allow the combination of the products to regenerate the original reactants. In batteries, this allows the creation of stored potential in a redox cell that will be released and harnessed to power a device. During discharge (when the battery is supplying electrical energy), the anode (negative electrode) undergoes oxidation (loses electrons) while the cathode (positive electrode) undergoes reduction (gains electrons). During charging (when the battery is being recharged), the electrodes are reversed to allow the restoration of the original components of the cell.

Fuel cells

Fuel cells (see section R1.3.5 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/fuel-cells-id-44783/)) generate electrical energy directly from a redox reaction. The fuel, such as hydrogen or methanol, is oxidised at the anode, generating electrons and cations.

For the hydrogen fuel cells in acidic conditions, the net reaction is:

$$2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$$

and can be divided into two half-reactions.

Oxidation:

$$H_2(g) \rightarrow 2H^+(aq) + 2e^- \quad \text{loss of electrons}$$

Reduction:

$$\tfrac{1}{2}O_2(g) + 2H^+(aq) + 2e^- \rightarrow H_2O(l) \quad \text{gain of electrons}$$

The electrons pass through an external circuit, producing an electrical current, while the cations move through the electrolyte to the cathode. At the cathode, the electrons combine with oxygen, or another oxidising agent, to form water or another byproduct. The continuous flow of electrons through the external wire generates a steady supply of electrical energy.

For the methanol fuel cells, the combustion of methanol in the presence of oxygen gas produces carbon dioxide and water vapour, according to the equation:

$$2CH_3OH(g) + 3O_2(g) \rightarrow 2CO_2(g) + 4H_2O(g)$$

A methanol fuel cell works with the same chemical reaction but separates the process into two half-reactions.

Oxidation:

$$CH_3OH(g) + H_2O(l) \rightarrow CO_2(g) + 6H^+(aq) + 6e^-$$

Reduction:

$$\tfrac{1}{2}O_2(g) + 2H^+(aq) + 2e^- \rightarrow H_2O(l)$$

Nature of Science

Aspect: Science as a shared endeavour

The development of hydrogen fuel cells may be the solution to the use of fossil fuels in motor vehicles. How can scientists work together to overcome global problems such as these

Advantages and disadvantages of the different cells

It is now time to consider the advantages and disadvantages of the different types of cells you have studied. These are summarised in **Table 1**.

Table 1. Advantages and disadvantages of the different typ

	Primary cells	**Secondary cells**
Distinctive feature	Single-use battery Irreversible reaction	Rechargeable battery Reversible reaction
Advantages	Longer shelf life Lower cost Greater reliability	Higher energy density Longer lifespan Lower environmental impact
Disadvantages	Lower energy densities Generate waste when discarded Environmental impact	Lead-acid batteries — heavy and require regular maintenance Nickel-cadmium (Ni-Cd) batteries — contain toxic cadmium, harmful to the environment Nickel-metal hydride (Ni-MH) batteries — less durable and have a shorter lifespan Lithium-ion (Li-ion) batteries — can be expensive and require careful handling to avoid safety risks

	Primary cells	Secondary cells	
Uses	Medical devices Emergency equipment Remote sensors	Lead-acid batteries — commonly used in automobiles Nickel-cadmium (Ni-Cd) batteries — from portable electronics to aircraft Nickel-metal hydride (Ni-MH) batteries — consumer electronics and hybrid electric vehicles Lithium-ion (Li-ion) batteries — portable electronics, electric vehicles, and renewable energy systems	f

Use **Interactive 5** to identify some advantages of different cells.

Use	Desirable property	Type of cell
	Reliability	Primary cells
Electric vehicles	High energy density	
	Continuous reaction	Fuel cells

Back-up power generation

Secondary cells

Fire alarm

Check

Kog

Interactive 5. Advantages of different cells.

Making connections

Electrical energy can be derived from the combustion of fossil fuels or from electrochemical reactions. What are the similarities and differences in these reactions? (see subtopic R1.3 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43475/))

Similarities

Electrical energy is an essential part of modern life. To produce it, we use fossil fuels, such as coal, natural gas, oil and electrochemical reactions, such as hydrogen fuel cells and lithium-ion batteries. Both the combustion of fossil fuels and electrochemical reactions convert chemical energy into electrical energy. To harness or use the electricity produced, both methods require the use of an external circuit to allow for the flow of electrons, which generates an electrical current.

Differences

Fossil fuel combustion is a non-renewable process, while some electrochemical reactions in batteries are a renewable process, as they can be recharged and reused.

Electrochemical reactions in batteries are often used for smaller-scale applications, such as portable electronics and electric vehicles, while fossil fuel combustion typically involves a large-scale power generation process.

The combustion of fossil fuels releases harmful emissions into the environment, including carbon dioxide, sulfur dioxide, and nitrogen oxides, which contribute to climate change and air pollution, while electrochemical reactions in batteries do not produce any harmful emissions.

Fossil fuels have historically been cheaper to obtain than the production of batteries, although the cost of batteries is decreasing as technology improves and demand increases.

Electrochemical reactions in batteries are more energy efficient processes, as some of the energy is lost as heat during the combustion process of fossil fuels.

Activity

- **IB learner profile attribute:** Inquirer
- **Approaches to learning:** Social skills — Assigning and accepting specific roles during group activities

Kog

- **Time required to complete activity:** 30 minutes
- **Activity type:** Pair/group activity

1. Use this simulation (https://teachchemistry.org/classroom-resources/galvanic-voltaic-cells-2) to create your own primary cells by selecting the electrode pairs and their molarities, and answer the following questions:

   - Find the combination of electrodes and their molarities that produces the highest electrical output.
   - Predict what will happen at each electrode and the salt bridge and click the 'see molecular scale buttons' to check if you were correct.
   - Write the half-reaction happening at each electrode.

2. With your pair or group, research other types of rechargeable batteries and write their charging and discharging equations. In your research, include common uses and the benefits/drawback of using this type of battery.